

DR2500D Rear Helper Spring Kit Installation Manual

Model: DR2500D (Rear Suspension Package)

Please read these instructions carefully before starting the installation. Ensure all standard safety precautions are met while working underneath the vehicle.

1. Parts List

Before beginning, verify that all components listed in the table below are present in your kit.

Ref #	Description / Specifications	Qty
1	Weldment Bracket (Left-Hand & Right-Hand)	1L - 1R
2	Rubber Spacer (4" O.D. x 1" Thick with AH Bushing)	2
3	Hollow Rubber Spring	2
4	Hex Cap Screw (HCS) (M10 x 40 x 1.5, Grade 8.8, Zinc Plated)	2
5	Hex Cap Screw (HCS) (M10 x 65 x 1.5, Grade 8.8, Zinc Plated)	2

2. Installation Instructions

- 1. Raise Vehicle:** Elevate the vehicle using an appropriate lifting device, allowing the rear axle to hang freely. Ensure the chassis is securely supported before proceeding.
- 2. Remove Bump Stop:** Locate and remove the existing OEM bump stop from the vehicle frame.
- 3. Assemble Components:** Assemble the hollow rubber spring (Ref 3) with the weldment bracket (Ref 1). Refer to the provided diagrams for correct orientation.
- 4. Mount Assembly:** Secure the assembled helper spring unit to the vehicle frame using the original equipment manufacturer (OEM) fasteners.
- 5. Lower Vehicle:** Carefully lower the vehicle back to the ground.

Installation Notes:

- To achieve a desired ride height, the rubber spacer (Ref 2) and the longer hex cap screw (Ref 5) may be utilized during assembly.
- When removing OEM fasteners, the application of penetrating fluid is highly recommended to facilitate removal.

3. Clearance Instructions & Fitting Guidance

- **Clearance Gap:** For rear helper spring kits installed on light-duty trucks and vans, an optimal clearance of 1/2" to 1-1/2" should be maintained between the rubber spring and its contact surface when the vehicle is unladen.
- **Load Effects:** The presence of a partial load, cargo, or aftermarket vehicle accessories will reduce or potentially eliminate this specified clearance gap.
- **Front Kit Guidance:** Front helper spring kits typically maintain constant contact with their respective engagement points.
- **Heavy-Duty Applications:** For medium and heavy-duty trucks, trailers, and motorhomes, the helper springs are designed to be in constant contact, thus exhibiting no clearance gap.

Important Safety Warnings:

- Do not exceed the vehicle manufacturer's recommended axle weight rating.
- Check for potential interference between the hollow rubber spring and other vehicle components. Under load, the rubber spring will expand in diameter.

Note: Illustrations are for reference only. If no specific reference image is provided, please install according to the standard mechanical guidelines.